

moves on the detector, as shown in Fig. 9.

Detector signal is suitably processed to determine relative distance to the target. While CMOS and CCD sensors detect peak distribution of light quantity on a sensor pixel array to identify target position, PSD sensors calculate beam centroid based on the entire reflected spot on an array.

The sensor is interfaced to the PC. Multiple sensors may be used when the objective is to determine displacement produced in a metal surface or a structure under stress due to blast loading or other similar effects. These are connected through an Ethernet interface card. Fig. 10 shows the schematic of a typical multi-sensor instrumentation.

CCD and CMOS systems are typi-

cally more accurate over a wider variety of surfaces, because only the highest charged pixels from the reflected beam are used to calculate position. Lower charged pixels are usually energised by unwanted reflections from changing optical properties of the surface being measured and can easily be ignored during signal processing. This allows these to be used in a wider variety of applications.

COTS laser triangulation sensors with a wide range of specifications of ranging distance, distance accuracy and range update rate are available from many manufacturers. Some representative modules from major international manufacturers include MICROTRAK-3 from MTI Instruments (Fig. 11), ILD-2300-200 from Micro-Epsilon, HL-G1 from Panasonic (Fig. 12), LDE-HS series from FAE (Fig. 13) and IL-2000 from Keyence.

These laser triangulation sensors represent a wide range of performance specifications in terms of stand-off range, measurement resolution and sampling rate. Triangulation sensors with stand-off range approaching a few metres, resolution approaching 0.01 microns and sampling rate of the order of a few hundred kHz are commercially available for a variety of industrial applications.

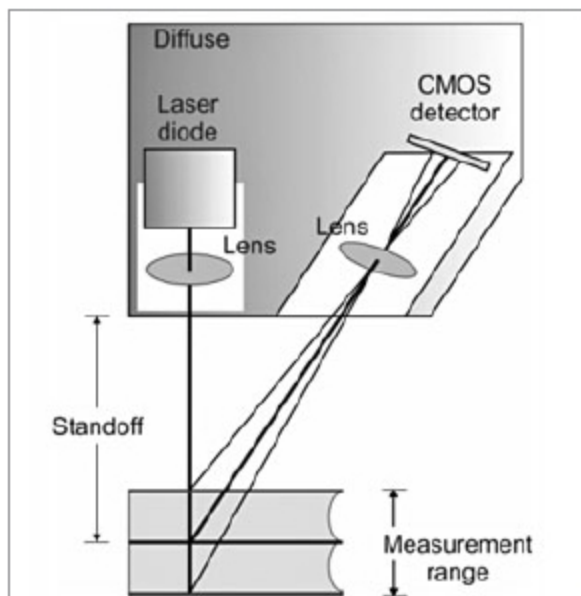


Fig. 9: Laser beam proportionally moving on detector



Fig. 11: MICROTRAK-3 from MTI Instruments

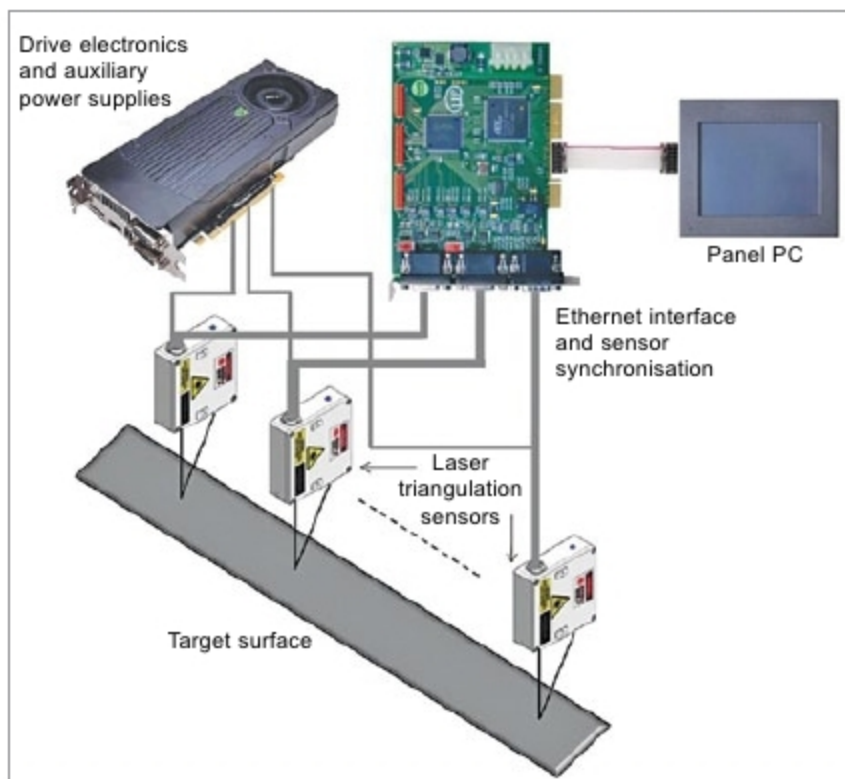


Fig. 10: Schematic of a typical multi-sensor instrumentation



Fig. 12: HL-G1 from Panasonic



Fig. 13: LDE-HS series from FAE

Laser velocity screen

A laser velocity screen is laser instrumentation comprising a two-dimensional array of optronic transmitter and receiver pairs. It is used for the measurement of velocity of projectiles of different calibres and travel velocities.

In addition to measuring projectile velocity, the instrumentation setup also measures aiming accuracy through the measurement of x-y coordinates of the projectile at two different planes that are spatially apart and parallel to each other, and orthogonal to the direction of travel of the projectile.

A two-dimensional laser screen may be created either by using separate transmitter and receiver pairs as shown in Fig. 14, or by using a laser transceiver unit on one side and a retro-reflector on the other.

Yet another approach could be to employ a single transmitter, a single receiver and multiple retro-reflectors for each plane, as shown in Fig. 15. With laser screens of this type, one cannot determine coordinates and, hence, aiming accuracy of the projectile. Fig. 16 shows the block schematic arrangement of different sub-systems comprising laser velocity screen instrumentation.

The integrated instrumentation functions as follows. Two screens are mounted parallel to each other, and spaced a known distance apart. In the case of laser screens using separate transmitter-receiver pairs, each of the two screens has two sets of linear semiconductor laser diode arrays located at right angles to each other in the same plane. Each linear array is aligned to a corresponding linear array of silicon PIN photodiodes. Assuming Cartesian coordinate system and that the centre of screen is the origin, we get two sets of collimated laser beams, one parallel to x-axis from linear arrays of laser diodes and photodiodes placed along y-axis, and the other parallel to y-axis obtained from linear arrays of laser diodes and photodiodes placed along x-axis.

The number of laser diode modules in the linear array, which is equal to the number of photodiodes, and the size of collimated laser beam are so chosen that at least one laser beam is interrupted by the smallest calibre projectile passing through it. As the projectile